

FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION

Revision Test 2026 — ANSWER KEY (Version 1)

Mathematics — Class 9 | Exam-15 | Total Marks: 43

Syllabus: Unit 1 — Real Numbers | Unit 2 — Logarithms | Unit 4 — Factorization & Algebraic Manipulation | Unit 5 — Linear Equations & Inequalities | Unit 7 — Coordinate Geometry

Every question on this paper is traceable to Seemab's own textbook. Each MCQ below carries the exact Student Learning Outcome (SLO) or section it comes from. No question uses general knowledge, no question comes from another subject's book, and nothing falls outside the studied syllabus.

SECTION A — MCQ Answers ($24 \times 1 = 24$ Marks)

1. The number $\sqrt{2}$ is: → **B. an irrational number** 1

$\sqrt{2}$ cannot be written as p/q , and its decimal neither terminates nor recurs, so it is **irrational**.

Syllabus reference: U1 SLO: real numbers as union of rationals and irrationals

2. In the radical $\sqrt[3]{8}$, the number 8 is called the: → **B. radicand** 1

The number under the radical sign is the **radicand**; the small 3 is the index.

Syllabus reference: U1 SLO: concept of radicals and radicands

3. The exponential form of $\sqrt[5]{(x^3)}$ is: → **A. $x^{3/5}$** 1

$\sqrt[n]{(x^m)} = x^{m/n}$, so $\sqrt[5]{(x^3)} = x^{3/5}$.

Syllabus reference: U1 SLO: transform radical form to exponential form

4. 0.333... is an example of a: → **B. non-terminating recurring decimal** 1

The digit 3 repeats forever, so it is **non-terminating but recurring**, and therefore rational.

Syllabus reference: U1 SLO: terminating and non-terminating recurring decimals

5. Using the laws of exponents, $a^5 \times a^{-2}$ equals: → **B. a^3** 1

When multiplying like bases the exponents add: $5 + (-2) = 3$, giving a^3 .

Syllabus reference: U1 SLO: apply the laws of exponents

6. Written in standard form of scientific notation, 45000 is: → **B. 4.5×10^4** 1

Standard form needs one non-zero digit before the decimal point: 4.5×10^4 .

Syllabus reference: U2 SLO: express a number in standard form

7. The value of $\log_2 8$ is: → **B. 3** 1

$\log_2 8$ asks: 2 to what power gives 8? Since $2^3 = 8$, the answer is 3.

Syllabus reference: U2 SLO: define logarithm of a number to the base a

8. The integral part of a common logarithm is called the: → **B. characteristic** 1

The whole-number part is the **characteristic**; the decimal part is the mantissa.

Syllabus reference: U2 SLO: characteristic and mantissa

9. A logarithm taken to the base e is called a: → **B. natural logarithm** 1

Base 10 gives the common logarithm; base e gives the **natural logarithm**.

Syllabus reference: U2 SLO: differentiate between common and natural logarithm

10. According to the laws of logarithms, $\log(mn)$ equals: → **B. $\log m + \log n$** 1

The product law: the log of a product is the **sum** of the logs.

Syllabus reference: U2 SLO: prove and apply the four basic laws of logarithm

11. The factors of $a^2 - b^2$ are: → **B. $(a + b)(a - b)$** 1

Difference of two squares: $a^2 - b^2 = (a + b)(a - b)$.

Syllabus reference: U4 SLO: recall factorization of $a^2 - b^2$

12. The expression $x^2 + 5x + 6$ factorizes to: → **A. $(x + 2)(x + 3)$** **1**

Two numbers multiplying to 6 and adding to 5 are 2 and 3, giving **$(x + 2)(x + 3)$** .

Syllabus reference: U4 SLO: factorize expressions of the type $x^2 + px + q$

13. The expression $a^3 + b^3$ factorizes to: → **A. $(a + b)(a^2 - ab + b^2)$** **1**

Sum of cubes: $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$. Note the minus sign in the middle term.

Syllabus reference: U4 SLO: factorize $a^3 + b^3$

14. For any two algebraic expressions p and q, the product $HCF \times LCM$ equals: → **B. $p \times q$** **1**

The product of the HCF and the LCM of two expressions equals the **product of the expressions**.

Syllabus reference: U4 SLO: know the relationship between HCF and LCM

15. Factorizing $ka + kb + kc$ gives: → **A. $k(a + b + c)$** **1**

Take the common factor k outside the bracket: **$k(a + b + c)$** .

Syllabus reference: U4 SLO: recall factorization of $ka + kb + kc$

16. The expression $a^2 + 2ab + b^2$ is the expansion of: → **B. $(a + b)^2$** **1**

It is a perfect square: $a^2 + 2ab + b^2 = (a + b)^2$.

Syllabus reference: U4 SLO: recall factorization of $a^2 + 2ab + b^2$

17. The value of $|-7|$ is: → **B. 7** **1**

Absolute value is the distance from zero and is never negative, so $|-7| = 7$.

Syllabus reference: U5 SLO: define absolute value

18. The solution of the equation $4x - 12 = 0$ is: → **A. $x = 3$** **1**

$4x = 12$, so $x = 12 \div 4 = 3$.

Syllabus reference: U5 SLO: solve linear equation with rational coefficients

19. The property stating that for any real numbers a and b exactly one of $a < b$, $a = b$, $a > b$ is true is called the: → **C. trichotomy property** **1**

The **trichotomy** property: exactly one of the three relations holds.

Syllabus reference: U5 SLO: recognize properties of inequalities

20. Dividing both sides of the inequality $-2x < 6$ by -2 gives: → **B. $x > -3$** **1**

Dividing an inequality by a **negative** number reverses the sign, giving **$x > -3$** .

Syllabus reference: U5 SLO: properties of inequalities (multiplicative); solve linear inequalities

21. If a and b are the coordinates of two points A and B on a coordinate line, the distance $d(A, B)$ is: → **B. $|b - a|$** **1**

$d(A, B) = |b - a|$. The absolute value makes the order of subtraction unimportant.

Syllabus reference: U7: Distance Between Points on a Coordinate Line

22. The distance between the points (2, 1) and (6, 4) is: → **A. 5 units** **1**

$d = \sqrt{(6-2)^2 + (4-1)^2} = \sqrt{16 + 9} = \sqrt{25} = 5$ **units**.

Syllabus reference: U7 SLO: find distance between two points in the plane

23. The mid-point of the line segment joining $(-3, 5)$ and $(7, 1)$ is: → **A. (2, 3)** **1**

$M = ((-3+7)/2, (5+1)/2) = (4/2, 6/2) = (2, 3)$.

Syllabus reference: U7 SLO: derive mid-point formula and calculate mid-point

24. Three or more points that lie on the same straight line are said to be: → **B. collinear** **1**

Points on one straight line are **collinear**; otherwise they are non-collinear.

Syllabus reference: U7 SLO: represent and identify collinear and non-collinear points

SECTION B — Short Question Answers ($3 \times 3 = 9$ Marks)

Q.2 (i) Exponential form and the laws of exponents. **3**

(a) Using $\sqrt[n]{x^m} = x^{m/n}$: **2**

(i) $\sqrt[4]{x^5} = x^{5/4}$ (ii) $\sqrt[3]{a^2} = a^{2/3}$

(b) **1**

$$(x^3 \times x^5) \div x^4 = x^{3+5} \div x^4 = x^8 \div x^4 = x^{8-4} = x^4$$

Syllabus reference: Unit 1 SLOs "transform an expression given in radical form to an exponential form" and "apply the laws of exponents to simplify expressions".

Q.2 (ii) Laws of logarithms. **3**

(a) **1**

$$\log m + \log n - \log p = \log(mn) - \log p = \mathbf{\log(mn / p)}$$

(product law for the addition, quotient law for the subtraction)

(b) **2**

Let $\log_5 125 = x$, so $5^x = 125$

$$125 = 5 \times 5 \times 5 = 5^3$$

So $5^x = 5^3$, giving $x = 3$

Syllabus reference: Unit 2 SLOs "define logarithm of a number to the base a" and "prove and apply the four basic laws of logarithm".

Q.2 (iii) Solving a linear inequality. **3**

$$7x - 4 \leq 24$$

Add 4 to both sides (**additive property of inequality**) **1**

$$7x \leq 28$$

Divide both sides by 7 (**multiplicative property**; 7 is **positive**, so the inequality sign does **not** reverse)

$$\mathbf{1}$$

$$\mathbf{x \leq 4}$$

Solution set = { x | x ≤ 4, x ∈ R } **1**

Syllabus reference: Unit 5 SLOs "recognize properties of inequalities" and "solve linear inequalities with rational coefficients".

SECTION C — Long Question Answers (2 × 5 = 10 Marks)

Q.3 (i) Factorization, HCF and LCM. **5**

(a) $x^2 + 9x + 20$ **1**

Two numbers multiplying to 20 and adding to 9 are 4 and 5.

$$x^2 + 9x + 20 = \mathbf{(x + 4)(x + 5)}$$

(b) $8a^3 + 27b^3$ **2**

Write as a sum of two cubes: $8a^3 = (2a)^3$ and $27b^3 = (3b)^3$

Using $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$:

$$= (2a + 3b)[(2a)^2 - (2a)(3b) + (3b)^2]$$

$$= \mathbf{(2a + 3b)(4a^2 - 6ab + 9b^2)}$$

(c) HCF and LCM of $8a^2b$ and $12ab^3$ **2**

$$8a^2b = 2^3 \times a^2 \times b \quad | \quad 12ab^3 = 2^2 \times 3 \times a \times b^3$$

$$\mathbf{HCF} = \text{lowest power of each common factor} = 2^2 \times a \times b = \mathbf{4ab}$$

$$\mathbf{LCM} = \text{highest power of every factor} = 2^3 \times 3 \times a^2 \times b^3 = \mathbf{24a^2b^3}$$

$$\mathbf{Verification:} \text{HCF} \times \text{LCM} = 4ab \times 24a^2b^3 = 96a^3b^4$$

$$\text{Product of the expressions} = 8a^2b \times 12ab^3 = 96a^3b^4 \quad \checkmark \text{ equal}$$

Syllabus reference: Unit 4 SLOs on factorizing $x^2 + px + q$ and $a^3 + b^3$, "find HCF and LCM of algebraic expressions", and "know the relationship between HCF and LCM".

Q.3 (ii) Distance and mid-point for A(-1, 2) and B(5, 10). **5**

(a) Distance AB **2**

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d = \sqrt{[(5 - (-1))]^2 + (10 - 2)^2}$$

$$d = \sqrt{[(6)^2 + (8)^2]} = \sqrt{(36 + 64)} = \sqrt{100}$$

AB = 10 units

(b) Mid-point M **2**

$$M = ((x_1 + x_2)/2 , (y_1 + y_2)/2)$$

$$M = ((-1 + 5)/2 , (2 + 10)/2) = (4/2 , 12/2)$$

M = (2, 6)

(c) Verification that M is equidistant from A and B **1**

$$AM = \sqrt{[(2 - (-1))]^2 + (6 - 2)^2} = \sqrt{(9 + 16)} = \sqrt{25} = 5 \text{ units}$$

$$MB = \sqrt{[(5 - 2)^2 + (10 - 6)^2]} = \sqrt{(9 + 16)} = \sqrt{25} = 5 \text{ units}$$

Since **AM = MB = 5 units**, M is equidistant from A and B, and $AM + MB = 10 = AB$. ✓

Syllabus reference: Unit 7 SLOs "find distance between two points in the plane", "derive mid-point formula and calculate mid-point of a line segment", and "apply distance and mid-point formulas".

— End of Answer Key —